DIVISIBILITY OF THE NUMERATORS OF $\sum_{k=1}^{36} k^{-n}$ BY 37^3: A PROOF OF ADAMCHUK'S CONJECTURES ON OEIS A116184

MATH0

Abstract. OEIS A116184 lists the integers $n \geq 1$ for which 37^3 divides the numerator of $H(n) = \sum_{k=1}^{36} k^{-n}$, and records two conjectures of Alexander Adamchuk (2007): every $n = 3 + 36k$ with $k \geq 0$ is a term, and every prime term $p > 37$ is of the form $37k - 1$. We prove both conjectures. The proof reduces the condition to the vanishing modulo 37^3 of a sum of n-th powers of 36 explicit residues, uses a binomial expansion truncated modulo 37^3 to show that it depends only on n mod 36 and $\lfloor n/36 \rfloor$ mod 37^2, and finishes with a finite computation. We found no earlier proof and classify the result as new, with the caveat that it is close to classical theory: the case $k = 0$ is a known consequence of 37 dividing the numerator of B_{32}, and the whole statement can also be derived from congruences of Sun, von Staudt–Clausen and Kummer (Remark 9). The theorem is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, reviewed for fidelity and frozen before the proof attempt, whose definitions are checked by the kernel against all 53 listed terms.

1. Introduction

For an integer $n \geq 0$ put

$$H(n) = \sum_{k=1}^{36} \frac{1}{k^n} \in \mathbb{Q},$$

the generalized harmonic number written $H(36, n)$ in the OEIS, and let $\operatorname{num} H(n)$ be its numerator in lowest terms. The sequence A116184 of the OEIS [11], contributed by Alexander Adamchuk on April 8, 2007, is

> "Numbers n such that 37^3 divides the numerator of generalized harmonic number H(36,n) = Sum[1/k^n, {k,1,36}]."

It has offset 1 and lists 53 terms, $3, 37, 39, 73, 75, 111, 147, 148, \ldots, 961, 975$. In the COMMENTS section Adamchuk wrote:

> "Conjecture: All terms of the arithmetic progression 3+36k belong to a(n). Prime terms in a(n) are {3, 37, 73, 443, 739, 887, 1109, ...}. It appears that all primes in a(n) that are greater than 37 are of the form 37k-1." (Alexander Adamchuk, Apr 08 2007)

We prove both conjectural claims (Theorem 1).

Background. The prime 37 is the smallest irregular prime: it divides the numerator of the Bernoulli number $B_{32} = -7709321041217/510$ and of no other B_e with $e \in \{2, 4, \ldots, 34\}$, so $(37, 32)$ is its only irregular pair (see [8] for irregular primes and Kummer's congruences). Wolstenholme-type congruences relate $H(p-1, m) = \sum_{k=1}^{p-1} k^{-m}$ modulo powers of p to Bernoulli numbers; see the survey of Meštrović [4]. For odd m with $1 \leq m \leq p - 4$, a congruence going

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back to Glaisher [2] and given in this form by Sun [7, Thm. 5.1(a)] (see also [3, Result 4]) states

$$H(p-1,m) \equiv \frac{m(m+1)}{2} \cdot \frac{B_{p-2-m}}{p-2-m}\, p^2 \pmod{p^3},$$

and Zhao [9, Thm. 2.8] shows that, for $p \geq 2n+3$, $p \mid B_{p-2n-1}$ if and only if $H(p-1,2n-1) \equiv 0 \pmod{p^3}$. With $p = 37$ and $m = 3$ this gives $37^3 \mid \operatorname{num} H(3)$, the case $k = 0$ of the first conjecture; Adamchuk recorded this divisibility in OEIS A007408 [10] in 2006. Zhao [9, §2.1] also studies the p-adic valuation of $H(p-1,s)$ for general s, via power sums and Kummer's congruences; he determines it for regular p and bounds it for irregular p under extra hypotheses, noting $p = 37$, $s = 1048$ as the first case where the bound fails without them [9, Rem. 2.6]. These results do not address the progression $3 + 36k$ or the prime terms.

Status. In the current revision of A116184 (#6, February 16, 2025) both statements are still conjectures, and the entry has no later comment on them. The math0 referees found no proof in the OEIS entries related to A116184 (A007408, A119722, A017533 and other entries by Adamchuk) or by web searches. We therefore classify Theorem 1 as a *new result*, with a caveat: it is elementary. Our proof is a periodicity argument plus a finite computation, and Remark 9 sketches an alternative derivation from classical congruences that a specialist could carry out in a few lines; we did not find that derivation written down.

2. Statement

Let $\mathcal{T} = \{n \in \mathbb{N} : \ 37^3 \mid \operatorname{num} H(n)\}$. Since $H(0) = 36$, $0 \notin \mathcal{T}$, so $\mathcal{T}$ is the set of terms of A116184.

Theorem 1.

(a) *For every integer $k \geq 0$, 37^3 divides the numerator of $H(3+36k)$.*
(b) *If $p > 37$ is a prime and 37^3 divides the numerator of $H(p)$, then $p = 37k-1$ for some integer $k \geq 1$, that is, $p \equiv -1 \pmod{37}$.*

The Lean statement. The statement was formalized by math0 in the file `lean/Math0/Statements/OEIS/A116184.lean`, reviewed for fidelity against the OEIS entry, and frozen (SHA-256 `a38dde38...f163f9`) before any proof attempt. The definitions used by the statement, the sequence `a`, and the statement itself are, verbatim:

```
/-- The generalized harmonic number ‘H(36, n) = ∑_{k=1}^{36} 1/k^n‘, computed exactly in ‘ℚ‘.
    -/
def H (n : ℕ) : ℚ := ∑ k ∈ Finset.Icc (1 : ℕ) 36, 1 / ((k : ℚ) ^ n)

/-- ‘n‘ is a term of A116184: ‘37^3‘ divides the numerator of ‘H(36, n)‘. -/
def IsTerm (n : ℕ) : Prop := (37 ^ 3 : ℤ) ∣ (H n).num

/-- A116184: the ‘n‘-th smallest (offset ‘1‘) number ‘m‘ such that ‘37^3‘ divides the numerator
of ‘H(36, m) = ∑_{k=1}^{36} 1/k^m‘. -/
noncomputable def a (n : ℕ) : ℕ := Nat.nth IsTerm (n - 1)

/-- Conjecture (Adamchuk, A116184): (a) every term ‘3 + 36k‘ (‘k ≥ 0‘) of the arithmetic
progression belongs to the sequence, and (b) every prime term greater than ‘37‘ is of the form
‘37k - 1‘. -/
def statement : Prop :=
  (∀ k : ℕ, IsTerm (3 + 36 * k)) ∧
    (∀ p : ℕ, p.Prime → 37 < p → IsTerm p → ∃ k : ℕ, p = 37 * k - 1)
```

All declarations are in the namespace `Math0.Statements.A116184`.

How the Lean definitions encode the entry. `H n` is $H(n)$, computed exactly in $\mathbb{Q}$, and `(H n).num` is Mathlib's `Rat.num`, the numerator in lowest terms. So `IsTerm n` is $n \in \mathcal{T}$, the defining property of the entry, and `a n = Nat.nth IsTerm (n - 1)` is the n-th smallest term with offset 1. The statement file proves that 0 is not a term (`not_isTerm_zero`) and that a positive integer x equals $a(m)$ for some $m \geq 1$ if and only if `IsTerm x` (`exists_a_eq_iff`); therefore "belongs to a(n)" and "primes in a(n)" are expressed through `IsTerm`. Part (a) of `statement` quantifies over $k \geq 0$; the progression starts at the listed term 3. In part (b), $37k-1$ is truncated subtraction in $\mathbb{N}$; it is harmless, because $k = 0$ gives $0 \neq p$, and for $k \geq 1$ it is ordinary subtraction. The list of prime terms quoted in the comment is example data, not a claim, and is not formalized.

The statement file also gives a decision procedure for `IsTerm` (Lemma 2 below; `isTerm_iff_check`) and a `DecidablePred` instance built from it. With it, the theorem `a_data` checks that $a(1), \ldots, a(53)$ are the 53 terms listed in the entry: the kernel evaluates a `List.filter` over $\{0, \ldots, 975\}$ (by `decide +kernel`), and a general lemma, `map_nth_eq_of_filter`, identifies the filtered list with the values `Nat.nth IsTerm i` for $i < 53$. (The entry's program scans $n \leq 1000$; the next term, $999 = 27 \cdot 37$, is not in the listed data and is not used.)

3. Proof

Throughout, $M = 37^3 = 50653$ and congruences without modulus are in $\mathbb{Z}/37^3\mathbb{Z}$. Let $\iota = 25327$; then $2\iota = M + 1$, so ι is the inverse of 2 in $\mathbb{Z}/37^3\mathbb{Z}$.

3.1. **Reduction to a sum of residues.** Let $F = 36!$ and, for $1 \leq k \leq 36$, let $c_k \in \{0, \ldots, M-1\}$ be the residue of F/k modulo M. The list $C = (c_1, \ldots, c_{36}) = (23531, 37092, 24728, \ldots, 18945)$ is written out in the statement file as `coeffs`, and `coeffs_eq` checks it by kernel evaluation. For $n \in \mathbb{N}$ let

$$\Sigma(n) = \sum_{k=1}^{36} c_k^n \in \mathbb{Z}/37^3\mathbb{Z}.$$

Lemma 2 (`H_eq`, `dvd_num_iff`, `isTerm_iff_check`, `check_iff`)**.** *For every $n \in \mathbb{N}$, $n \in \mathcal{T}$ if and only if $\Sigma(n) = 0$ in $\mathbb{Z}/37^3\mathbb{Z}$.*

Proof. Since $k \mid F$ for $1 \leq k \leq 36$, $H(n) = s/F^n$ with $s = \sum_{k=1}^{36}(F/k)^n \in \mathbb{N}$. As 37 is prime and $37 > 36$, $\gcd(M, F^n) = 1$. We claim that if $q = s/b$ with $b \neq 0$ and $\gcd(M, b) = 1$, then $M \mid \operatorname{num} q$ if and only if $M \mid s$. Indeed, writing $q = \operatorname{num} q / \operatorname{den} q$ in lowest terms, we have $\operatorname{num} q \cdot b = s \cdot \operatorname{den} q$ and $\operatorname{den} q \mid b$, so $\operatorname{den} q$ is prime to M as well. If $M \mid \operatorname{num} q$, then $M \mid s \operatorname{den} q$ and hence $M \mid s$; if $M \mid s$, then $M \mid \operatorname{num} q \cdot b$ and hence $M \mid \operatorname{num} q$. Finally $s \equiv \sum_k c_k^n \pmod{M}$. □

In the statement file this is `isTerm_iff_check`, with the Boolean test `check n`: $\big(\sum_{c \in C}(c^n \bmod M)\big) \bmod M = 0$; the proof file restates it in $\mathbb{Z}/37^3\mathbb{Z}$ (Mathlib's `ZMod 50653`) as `check_iff`.

3.2. **A truncated binomial expansion.** For $c \in C$ put $a_c = \big((c^{36} \bmod M) - 1\big)/37$ (integer division).

Lemma 3 (`A_spec`)**.** *For every $c \in C$, $c^{36} \bmod M = 1 + 37a_c$. In particular $c^{36} = 1 + 37a_c$ in $\mathbb{Z}/37^3\mathbb{Z}$, and $0 \leq a_c < 37^2$.*

Proof. Since $37 \nmid c$, $c^{36} \equiv 1 \pmod{37}$ by Fermat's little theorem, so $c^{36} \bmod M - 1$ is divisible by 37. In Lean the 36 instances are checked directly by kernel evaluation (`decide +kernel`); the values a_c lie between 7 and 1336. □

Lemma 4 (`key_pow`)**.** *Let $x, a \in \mathbb{Z}/37^3\mathbb{Z}$ with $x^{36} = 1 + 37a$. Then for all $r, t \in \mathbb{N}$,*

$$x^{r+36t} = x^r\big(1 + 37\,a\,t + 37^2\iota\,a^2\,t(t-1)\big).$$

Proof. As $x^{r+36t} = x^r(1+37a)^t$, it suffices to show $(1+37a)^t = Q(t)$, where $Q(t) = 1 + 37at + 37^2\iota a^2 t(t-1)$. For $t = 0$ both sides are 1. For the induction step, a direct expansion gives

$$Q(t)(1+37a) - Q(t+1) = 37^2a^2t\,(1-2\iota) + 37^3\iota a^3 t(t-1) = M\big(-37^2a^2t + \iota a^3 t(t-1)\big),$$

using $1 - 2\iota = -M$. This is 0 in $\mathbb{Z}/37^3\mathbb{Z}$, so $(1+37a)^{t+1} = Q(t)(1+37a) = Q(t+1)$. □

For $j \in \{0,1,2\}$ and $r \in \mathbb{N}$ define the natural numbers

$$S_j(r) = \sum_{c \in C} c^r a_c^j.$$

Lemma 5 (`sum_pow`)**.** *For all $r, t \in \mathbb{N}$, in $\mathbb{Z}/37^3\mathbb{Z}$,*

$$\Sigma(r + 36t) = S_0(r) + 37\,t\,S_1(r) + 37^2\iota\,t(t-1)\,S_2(r).$$

Proof. Apply Lemma 4 to each $c \in C$, which is allowed by Lemma 3, and sum over C. □

3.3. **Part (a).** By kernel evaluation, M divides $S_0(3)$, $37\,S_1(3)$ and $37^2 S_2(3)$; in fact $S_0(3) \equiv 0$, $S_1(3) \equiv 10952 = 8 \cdot 37^2$ and $S_2(3) \equiv 7215 = 195 \cdot 37 \pmod M$. Hence, by Lemma 5 with $r = 3$ and $t = k$,

$$\Sigma(3 + 36k) = S_0(3) + k \cdot \big(37 S_1(3)\big) + \iota k(k-1) \cdot \big(37^2 S_2(3)\big) = 0$$

for every $k \in \mathbb{N}$, and $3 + 36k \in \mathcal{T}$ by Lemma 2. This is `part_a`. □

3.4. **Part (b).** For $a_0, a_1, a_2 \in \mathbb{Z}/37^3\mathbb{Z}$ and $t \in \mathbb{N}$ let $E(t) = a_0 + 37ta_1 + 37^2\iota\,t(t-1)a_2$.

Lemma 6 (`reduce_t`)**.** *$E(t) = E(t \bmod 37^2)$ for every $t \in \mathbb{N}$.*

Proof. Write $t = u + 37^2 q$ with $u = t \bmod 37^2$. Then $37t = 37u + Mq$, and

$$t(t-1) = u(u-1) + 37^2 q\,(2u - 1 + 37^2 q),$$

so $37^2\iota\,t(t-1) - 37^2\iota\,u(u-1) = 37 \cdot M\iota q(2u - 1 + 37^2 q)$. Both differences vanish in $\mathbb{Z}/37^3\mathbb{Z}$. □

Thus, for fixed r, $\Sigma(r + 36t)$ depends only on $t \bmod 37^2$. The values $T_j(r) = S_j(r) \bmod M$ for $r < 36$ and $j = 0, 1, 2$ are written out as three lists of 36 numbers in the proof file (`T0`, `T1`, `T2`), and the kernel checks that they are correct (`T0_spec`, `T1_spec`, `T2_spec`). Let

$$R = \{r < 36 :\ r \text{ odd},\ 3 \nmid r\} = \{1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35\}.$$

Lemma 7 (`finite_b_aux`, `finite_b`)**.** *Let $r \in R$ and $0 \le u < 37^2$, and suppose $r + 36u \not\equiv 0 \pmod{37}$ and*

$$T_0(r) + 37\,u\,T_1(r) + 37^2\iota\,u\,(u + M - 1)\,T_2(r) \equiv 0 \pmod M.$$

Then $r + 36u \equiv 36 \pmod{37}$.

Proof. This is a finite check of $12 \cdot 1369 = 16428$ pairs (r, u). In Lean, the test for one pair is the Boolean function

```
private def okU (r a0 a1 a2 u : ℕ) : Bool :=
  (r + 36 * u) % 37 == 0 ||
    (a0 + 37 * u * a1 + 1369 * 25327 * u * (u + 50652) * a2) % 50653 != 0 ||
    (r + 36 * u) % 37 == 36
```

and `finite_b_aux` states that `okU r (T0.getD r 0) (T1.getD r 0) (T2.getD r 0) u = true` for all $r < 36$ with $r \bmod 2 = 1$, $r \bmod 3 \neq 0$, and all $u < 1369$. It is proved by `decide +kernel`: the kernel evaluates the bounded quantifiers and the test, using its built-in arithmetic on natural number literals. The factor $u + M - 1$ is used in place of $u - 1$ to avoid truncated subtraction; it is congruent to $u-1$ modulo M. (Exactly 407 pairs satisfy both hypotheses.) □

Proof of Theorem 1(b). Let $p > 37$ be a prime with $p \in \mathcal{T}$. Then p is divisible by none of 2, 3, 37. Write $p = r + 36t$ with $r = p \bmod 36$ and $t = \lfloor p/36 \rfloor$, and let $u = t \bmod 37^2$. Since $2 \mid 36$ and $3 \mid 36$, r is odd and $3 \nmid r$, so $r \in R$. By Lemmas 2, 5 and 6,

$$0 = \Sigma(p) = S_0(r) + 37\, u\, S_1(r) + 37^2 \iota\, u(u-1)\, S_2(r),$$

and reducing $S_j(r)$ modulo M gives the congruence in Lemma 7. Moreover $p = r + 36u + 36 \cdot 37^2 \lfloor t/37^2 \rfloor \equiv r + 36u \pmod{37}$, so $r + 36u \not\equiv 0 \pmod{37}$. Lemma 7 gives $r + 36u \equiv 36 \pmod{37}$, hence $p \equiv 36 \pmod{37}$ and $p = 37\big(\lfloor p/37 \rfloor + 1\big) - 1$. This is `part_b`. □

Theorem 1 is `result : statement`, proved as the pair of `part_a` and `part_b`.

Remark 8 (periodicity)**.** By Lemmas 2, 5 and 6, membership in $\mathcal{T}$ depends only on n mod $\varphi(37^3) = 36 \cdot 37^2 = 49284$. The proof organizes the check of one period so that part (a) needs three divisibilities and part (b) needs 16428 evaluations of a quadratic polynomial.

Remark 9 (a derivation from classical congruences; not formalized)**.** Theorem 1 also follows from classical results. Let $p = 37$, let n be odd, and choose N, a multiple of 49284, with $N > n + 4$. Then $H(n) \equiv \sum_{k=1}^{36} k^{N-n} \pmod{p^3}$ in $\mathbb{Z}_{(p)}$, and $m = N - n$ is odd with $m \geq 5$, so $B_m = B_{m-2} = 0$, and Sun's congruence for power sums [7] (see [3, Result 2]),

$$\sum_{k=1}^{p-1} k^m \equiv pB_m + \tfrac{p^2}{2} m B_{m-1} + \tfrac{p^3}{6} m(m-1) B_{m-2} \pmod{p^3},$$

gives $H(n) \equiv \frac{p^2}{2} m B_{m-1} \pmod{p^3}$. Note $m \equiv -n \pmod{p^2}$, since $p^2 \mid N$, and $m - 1 \equiv -(n+1)$ modulo p and modulo $p - 1 = 36$. If $n \not\equiv 35 \pmod{36}$, then $p - 1 \nmid m - 1$, so $B_{m-1}/(m-1)$ is p-integral, and by Kummer's congruences it is congruent modulo p to B_e/e, where $e \in \{2, \ldots, 34\}$ and $e \equiv -(n+1) \pmod{36}$. Since 32 is the only such index with $p \mid B_e$, and since $p \mid m \iff 37 \mid n$ and $p \mid m - 1 \iff 37 \mid n + 1$, one obtains: $n \in \mathcal{T}$ if and only if $37 \mid n$, or $37 \mid n + 1$, or $n \equiv 3 \pmod{36}$. If $n \equiv 35 \pmod{36}$, then $pB_{m-1} \equiv -1 \pmod{p}$ by the von Staudt–Clausen theorem, so $v_p(H(n)) = 1 + v_p(n)$ when $v_p(n) \leq 1$, and $n \in \mathcal{T}$ if and only if $37^2 \mid n$. Part (a) is the case $n \equiv 3 \pmod{36}$. For part (b), let $p' > 37$ be a prime in $\mathcal{T}$. Then p' is odd and $37^2 \nmid p'$, so $p' \not\equiv 35 \pmod{36}$ by the second case; and $37 \nmid p'$ and $p' \not\equiv 3 \pmod{36}$ (as $3 \nmid p'$), so $37 \mid p' + 1$ by the first case. We checked this characterization of the odd elements of $\mathcal{T}$ numerically over a full period $1 \leq n \leq 49284$; it is not part of the formal proof.

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A116184.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file (Section 2) and `Math0/Audit.lean`. The main declaration is `theorem result : statement` in the namespace `Math0.Frontier.A116184`; the file ends with the command

```
#math0_settles Math0.Frontier.A116184.result Math0.Statements.A116184.statement
```

Of the declarations named in Section 3, `coeffs_eq`, `H_eq`, `dvd_num_iff` and `isTerm_iff_check` are in the statement file; the others, except `result`, are private lemmas of the proof file. The algebra in $\mathbb{Z}/37^3\mathbb{Z}$ is done in `ZMod 50653`. Lemmas 4 and 6 are proved with `linear_combination`,

using the fact `(50653 : ZMod 50653) = 0`; the coefficients of these linear combinations are the multipliers of M displayed in the proofs above. Lemma 5 follows from Lemma 4 by `ring` and a list lemma, `sum_expand`, that distributes the sum.

The certificate. All finite facts are proved by `decide +kernel`, so they are checked by the Lean kernel, which evaluates arithmetic on natural number literals with its built-in big-number operations. They are: the residue table `coeffs_eq`, the data check `a_data` and `not_isTerm_zero` (statement file); `A_spec` (36 instances); the tables `T0_spec`, `T1_spec`, `T2_spec` (3×36 entries, involving c^r for $r \leq 35$); the three divisibilities for $r = 3$ in `part_a`; and `finite_b_aux` (16428 evaluations of `okU`). Neither file uses `native_decide`, `sorry`, `set_option` or new axioms. Checking the proof file with `lake env lean` takes about one minute.

Audit. The command `#math0_settles` (in `lean/Math0/Audit.lean`) is used for statements that math0 formalizes itself. It fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (so no `sorryAx`, no `native_decide`, and no custom axioms); and (3) the type of `result` is syntactically the constant `statement` or its negation, up to metadata, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A116184.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A116184.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A116184.result' proves 'Math0.Statements.A116184.statement'
```

Toolchain and reproduction. We use Lean 4 [6] `v4.33.1` and Mathlib [5] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use (A116184 is not in formal-conjectures). To reproduce, run `scripts/verify.sh` from the repository root; it builds all math0 results with `lake build`, re-running every audit command, and checks that the pinned dependencies are unmodified. The single file is checked by `lake env lean Math0/Frontier/OEIS/A116184.lean` in `lean/`.

References

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[10] The OEIS Foundation Inc., `https://oeis.org/A007408`, Wolstenholme numbers: numerator of $\sum_{k=1}^{n} 1/k^3$; comment by A. Adamchuk (Nov. 7, 2006).

[11] The OEIS Foundation Inc., `https://oeis.org/A116184`, entry by A. Adamchuk (Apr. 8, 2007); revision #6, Feb. 16, 2025.

Produced by Claude (an AI model developed by Anthropic) in the math0 project